Swetha M

Tiruppur, India swethadevi3567@gmail.com +91 9944256597 Swetha M

Summary

I am a passionate and enthusiastic Computer Science and Engineering student with strong technical skills, problem-solving abilities to excel in both independent and team environments, tackling challenges with resilience, creativity, and a commitment to continuous learning, innovation and contributing effectively to organizational goals.

Education

Sri Ramakrishna Engineering College — Coimbatore, TN, India
B.E Computer Science and Engineering; CGPA: 7.76 — *Nov 2022 – Present*

Kongu Vellalar Matric Higher Secondary School — Tiruppur, TN, India
Higher Secondary Education; Percentage: 92.83 % — *June 2021 – May 2022*

Projects

Kidney Condition Analysis Through Iris | *Python, OpenCV, Machine Learning, Iridiology*

- Developed a non-invasive kidney and cardiovascular health detection system using iridology, implementing a multi-stage image processing pipeline to classify normal vs. abnormal conditions from iris images.
- Applied thresholding-based classification by computing white-to-black pixel ratios within the ROI mapped from iridology charts, enabling cost-effective early screening without clinical diagnostic equipment

AI Customer Support Agent – LangChain RAG Chatbot | *Python, LangChain, FAISS, OpenAI API, Streamlit, RAG*

- Built a conversational AI support agent using LangChain and FAISS vector store to perform Retrieval-Augmented Generation over PDF knowledge bases, achieving 90% query accuracy and 40% faster retrieval than keyword search baseline.
- Integrated OpenAI API with conversation memory and tool-calling capabilities via a Streamlit UI, enabling context-aware multi-turn interactions over custom document collections.

A Real-Time Prompt Firewall for Securing LLM Applications | *Python, Flask, Ollama, Regex, RAG, SSE, JS, TailwindCSS*

- Built a hybrid 3-stage security pipeline (Regex Pattern Matching → RAG Similarity → Ollama LLM Classification) to detect and block jailbreak, injection, and harmful prompts in real time, with Morse code decoding to defeat encoding-based evasion attacks.
- Developed a Chrome Extension integrating with LLM platforms (ChatGPT, Claude, Gemini, Copilot) and a streaming secure chat UI with live pipeline visualization.

Skills

Languages	C, Python, Java, HTML, CSS, JavaScript, SQL
Tools	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Jira, Docker
Libraries/Frameworks	Scikit-learn, Flask, Pandas, NumPy, Streamlit, PyTorch, TensorFlow
Soft Skills	Problem Solving, Time Management, Team Collaboration, Highly Adaptable
Database	MySQL, MongoDB

Internship

Web Design Intern | Knila IT Solutions Pvt. Ltd

- Designed visually appealing and user-friendly interfaces using HTML, CSS, and UI/UX principles. Ensured responsive and consistent layouts across devices.
- Improved user experience by applying design best practices and conducting usability checks.

Web Development Intern | Twisys Infotech

- Developed and maintained responsive web applications using HTML, CSS, JavaScript. Focused on performance, responsiveness, and cross-browser compatibility.
- Integrated APIs, fixed bugs, and optimized performance to enhance functionality and user experience.

Certifications

Google Cloud Study Jams -- Google Cloud Platform | Getting Started with AI on Jetson Nano – NVIDIA | Cloud Computing Fundamentals -- IBM SkillsBuild | Cloud Foundations -- AWS academy | Privacy and Security in Online Social Media – NPTEL(IIIT, Hyderabad) | Cyber Security and Privacy – NPTEL(IIT, Madras)

Co-Curricular Activity

- Presented research paper at ICRCET-2024 (12th International Conference on Recent Challenges in Engineering & Technology), organised by IFERP.